



EAST WEST UNIVERSITY

Project Report
(EWU Lost & Found management System)

Course Title: Database Systems

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[TASS-nia/ewu lost found](#)

1. Project Description

For this project we built a Lost & Found Management System for East West University. The idea came from a pretty common problem on campus — people lose things like ID cards, wallets, phones, or umbrellas all the time, and right now there's no real system for it. Someone finds a wallet in the cafeteria, posts about it in a Facebook group or WhatsApp group chat, and hopes the owner sees it eventually. It works sometimes, but a lot of items just never get back to their owners because there's no organised record of what's been found or lost.

So the system we're simulating here is basically a digital version of a university Lost & Found desk. Instead of relying on posts and word of mouth, students can log in and report an item as either “lost” or “found”, browse everything that's currently been reported, and if they spot their own item in the found list, submit a claim explaining why it's theirs. An admin (standing in for the actual Lost & Found office staff) then checks these claims and approves or rejects them, which is what finally marks an item as returned.

In terms of data, the system mainly deals with four things: user accounts (students and admins), the item reports themselves (with details like category, location, date, and status), the categories used to organise those items, and the claims students submit when they think something belongs to them. The whole point is to keep track of an item's journey from the moment it's reported all the way to when it's actually handed back — and to have a proper record of who reported what and who verified it, instead of everything living in random group chats.

2. Key Features

We kept the feature set fairly focused since this is a course project, but it covers the full workflow from reporting an item to getting it back. There are two roles, Student and Admin, and each one can do different things once logged in.

Student Features:

- Sign up using their EWU student email — we added validation so only addresses ending in @std.ewubd.edu are accepted.
- Log in with a password (hashed in the database, never stored as plain text).
- Report something as Lost or Found, filling in a title, description, category, location, and date.
- Look through the full list of everything that's been reported so far.
- Submit a claim on an item they believe is theirs, along with a short explanation.
- Log out whenever they're done.

Admin Features:

- See every item report that's been submitted.
- Go through pending claims and approve or reject them — approving one automatically updates the item's status to “Claimed”.
- Add new categories or remove ones that aren't being used (the system won't let you delete a category that items are still linked to).
- Manage the user list — promote a student to Admin, demote an Admin back to Student, or delete an account, as long as that user hasn't already reported any items.

3. Database Design

3.1 E-R Model

The database is built around four entities user, category, item, and claim. A user (student or admin) can report many items and submit many claims on items reported by other users. Each item belongs to one category and can receive many claims from different students. Item table also has a self relationship which lets a 'lost' report linked to the matching 'found' report once admin confirms the two are same. Admin-related actions like approval and verification are included in the schema to support the complete workflow.

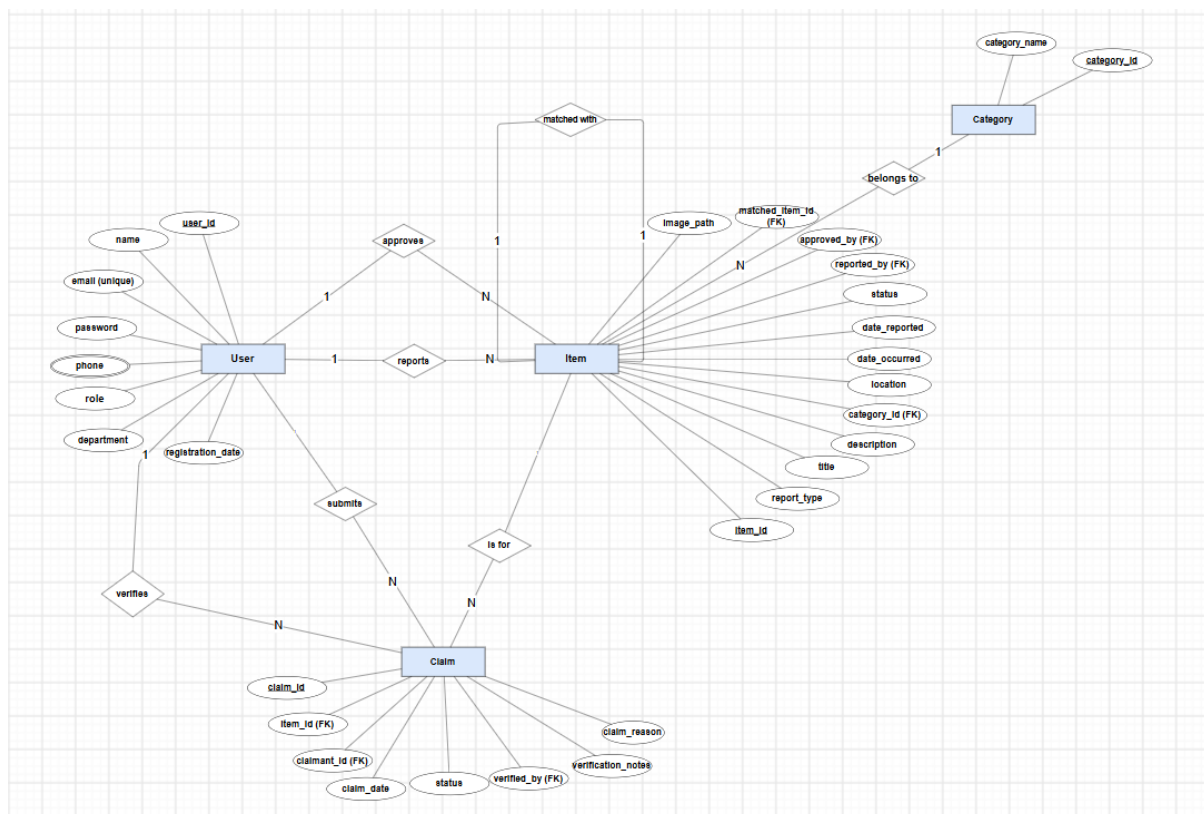


Figure 3.1 – ER Diagram

3.2 Relational Data Model

The ER model above translates into following four relational tables.

Table: User

Column	Type	Key / Notes
user_id	INT (Auto_Increment)	Primary Key
name	Varchar(100)	Not null
email	Varchar(150)	Unique; must end in @std.ewubd.edu
password	Varchar(255)	Stored as a bcrypt hash (password_hash)
phone	Varchar(20)	Optional
role	ENUM('Student','Admin')	Determines dashboard & access
department	Varchar(100)	Not null
registration_date	Date	Set to CURDATE() on registration

Table: Category

Column	Type	Key / Notes
category_id	INT (Auto_Increment)	Primary Key
category_name	Varchar(100)	Not null

Table: Item

Column	Type	Key / Notes
item_id	INT (Auto_Increment)	Primary Key
report_type	ENUM('Lost', 'Found')	Type of report filed
title	Varchar(150)	Short name or title of item
description	Text	Detailed description of item
category_id	INT	Foreign Key → Category(category_id)
location	Varchar(200)	Place where item was lost/found
date_occurred	Date	Date event occurred
date_reported	Date	Set to CURDATE() on submission
status	ENUM('Pending', 'Approved', 'Rejected')	Default 'Pending'
reported_by	INT	Foreign Key → User(user_id)
approved_by	INT	Foreign Key → User(user_id); admin who decided
matched_item_id	INT	Foreign Key → Item(item_id); optional linked item
image_path	Varchar(255)	Image file path/URL

Table: Claim

Column	Type	Key / Notes
claim_id	INT (Auto_Increment)	Primary Key
item_id	INT	Foreign Key → Item(item_id)
claimant_id	INT	Foreign Key → User(user_id)
claim_reason	Text	Student's proof / explanation
claim_date	Date	Set to CURDATE() on submission
status	ENUM('Pending', 'Approved', 'Rejected')	Default 'Pending'
verified_by	INT	Foreign Key → User(user_id); admin who decided
verification_notes	Text	Optional admin notes

Relational Schema Diagram:

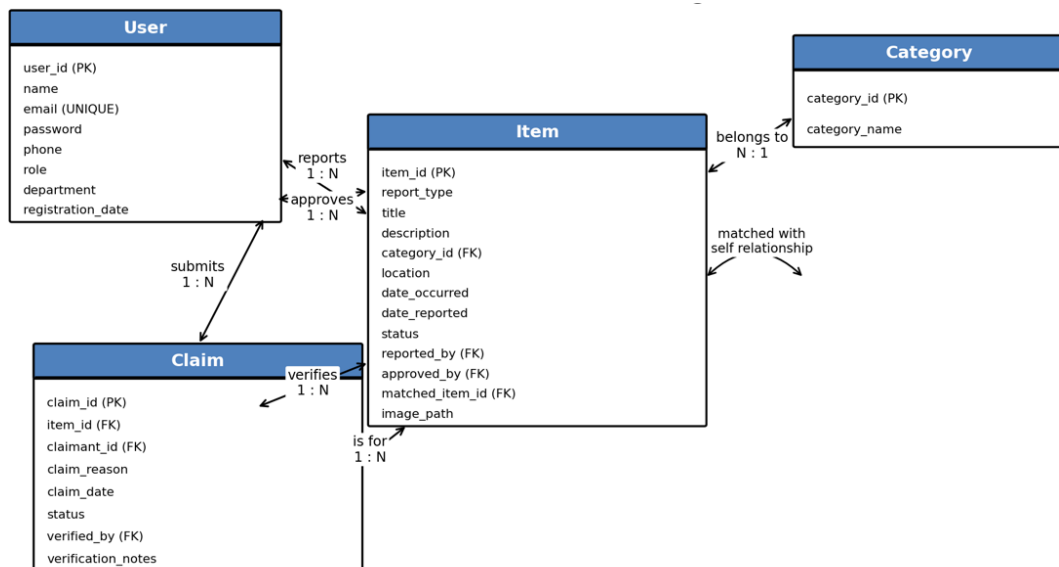


Figure 3.2 –Relational Schema Diagram

Relational Schema Table:

Entity 1	Relationship	Entity 2	Cardinality	Meaning
User	reports	Item	1 : N	One user can report many items, each item is reported by one user.
User	approves	Item	1 : N	One user(Admin) can approve many items, each item is approved by one user(Admin).
Category	belongs to	Item	1 : N	One category can contain many items, each item belongs to one category.
Item	matched with	Item	1 : 1	An item can be matched with one other item and it is one-to-one relation.
Item	is for	Claim	1 : N	One item can have many claims, each claim is for one item.
User	submits	Claim	1 : N	One user can submit many claims, each claim is submitted by one user.
User	verifies	Claim	1 : N	One user(Admin) can verify many claims, each claim is verified by one user(Admin).

4. Tools and Technologies Used

The EWU Lost & Found Management System was developed using the following tools and technologies:

- **PHP:** Used for server-side programming and implementing the application logic.
- **MySQL:** Used as the relational database management system for storing and managing system data.
- **SQL:** Used to create, retrieve, update, and manage data in the database.

- **HTML:** Used to design the structure and user interface of the web pages, including forms and tables.
- **CSS:** Used for styling web pages, including layout, colours, badges, and form elements to create an attractive user interface.
- **XAMPP:** Used as the local development environment for running Apache, PHP, and MySQL.
- **VS Code:** Used as the primary code editor for writing and managing the project files.
- **draw.io:** Used to design and create the Entity-Relationship (ER) diagram and database schema diagram.
- **Windows:** Used as the operating system during the development of the project.

5. Role Assignment

The system has two types of users: **Student** and **Admin**. Their roles are stored in the User table and checked using PHP sessions.

Student

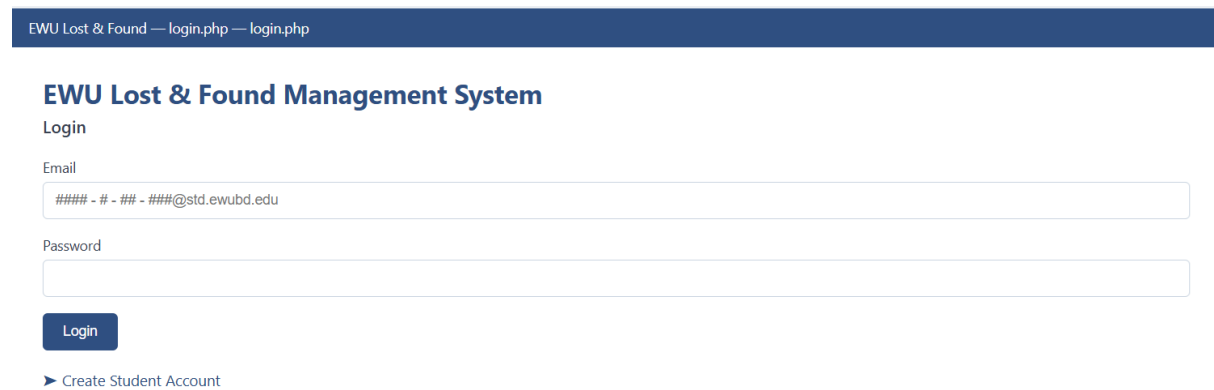
- A new user is automatically registered as a Student.
- Students can report lost or found items.
- They can view all reported items.
- They can submit claims for available items.
- They cannot access admin pages. If they try, they are redirected to the login page.

Admin

- An Admin can be created through the setup system or by promoting an existing Student.
- Admins can access the Admin Dashboard.
- They can approve or reject claims and update the item status.
- They can add or delete item categories.
- They can promote or demote users and delete users who have no related item reports.

6. GUI Screenshot

Login Page: Students and admins enter their EWU email and password to access the system. New students can also create student account.

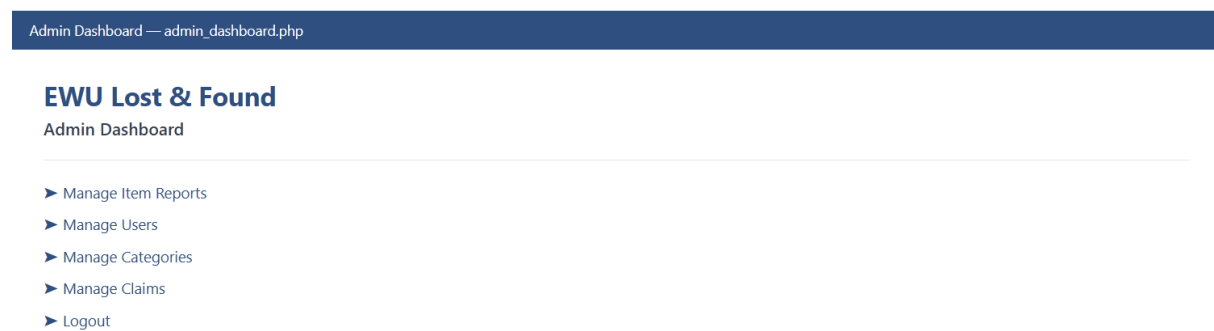


The screenshot shows the login page of the 'EWU Lost & Found Management System'. At the top, a dark blue header bar contains the text 'EWU Lost & Found — login.php — login.php'. Below the header, the title 'EWU Lost & Found Management System' is displayed in a bold, dark blue font. Underneath the title, the word 'Login' is written in a smaller, dark blue font. The page features two input fields: 'Email' and 'Password'. The 'Email' field contains a placeholder text '#### - # - ## - ###@std.ewubd.edu'. Below the 'Password' field, there is a dark blue 'Login' button. At the bottom of the form, there is a link that says '► Create Student Account'.

Figure 6.1 - Login page (login.php)

(Admin Side)

Admin Dashboard: Main menu for admin or admins to manage item reports, users, categories, and claims.



The screenshot shows the 'Admin Dashboard' of the 'EWU Lost & Found Management System'. At the top, a dark blue header bar contains the text 'Admin Dashboard — admin_dashboard.php'. Below the header, the title 'EWU Lost & Found' is displayed in a bold, dark blue font. Underneath the title, the text 'Admin Dashboard' is written in a smaller, dark blue font. The page features a list of links, each preceded by a right-pointing arrow: '► Manage Item Reports', '► Manage Users', '► Manage Categories', '► Manage Claims', and '► Logout'.

Figure 6.2 - Admin Dashboard (admin_dashboard.php)

View Items: Displays all reported lost and found items with status pending, approved, claimed, closed.

View Items — view_items.php								
EWU Lost & Found								
All Reported Items								
Type	Title	Category	Location	Date Occurred	Date Reported	Status	Reported By	Description
Lost	Chemistry Textbook	Accessories	CSE Building	2026-08-13	2026-08-19	Pending	Mim Akter	Organic Chemistry textbook, 3rd edition, name written inside
Lost	Chemistry Textbook	Accessories	CSE Building	2026-08-13	2026-08-19	Approved	Faisal Ahmed	Organic Chemistry textbook 3rd edition
Lost	Blue Earbuds	Bags	Library 3rd Floor	2026-08-15	2026-08-19	Approved	Sadia Islam	Apple AirPods Pro, white color, in charging case
Lost	Blue Earbuds	Bags	Library	2026-08-15	2026-08-19	Approved	Faisal Ahmed	Apple AirPods Pro, white color
Found	Blue Umbrella	Documents	Main Gate	2026-08-17	2026-08-19	Pending	Shohanur Alam Shovon	Navy blue umbrella with wooden handle found near main gate
Found	Red Scarf	Documents	Sports Ground	2026-08-14	2026-08-19	Approved	Tanvir Ahmed	Red wool scarf, possibly from the winter collection
Found	Blue Umbrella	Documents	Main Gate	2026-08-17	2026-08-19	Approved	Faisal Ahmed	Navy blue umbrella with wooden handle
Found	Red Scarf	Documents	Sports Ground	2026-08-14	2026-08-19	Approved	Faisal Ahmed	Red wool scarf

Figure 6.3 - View Lost & Found Items (view_items.php).

Manage Users: Shows all registered users with options to change roles (Admin <> Student) or delete accounts.

Manage Users — manage_users.php							
EWU Lost & Found							
Manage Users							
ID	Name	Email	Phone	Role	Department	Registered On	Action
3	Nusrat Jahan Mridha	2025-1-60-198@std.ewubd.edu	01927815825	Admin	CSE	2026-08-15	Make Student Delete
4	Shohanur Alam Shovon	2023-1-60-271@std.ewubd.edu	01892294691	Student	EEE	2023-01-15	Make Admin Delete
5	Sadia Islam	2020-2-60-102@std.ewubd.edu	01822334455	Student	BBA	2026-01-20	Make Admin Delete
6	Tanvir Ahmed	2019-1-60-089@std.ewubd.edu	01933445566	Student	EEE	2026-02-01	Make Admin Delete
7	Mim Akter	2021-3-60-015@std.ewubd.edu	01644556677	Student	CSE	2026-02-10	Make Admin Delete
10	Alina Khan	2023-1-60-045@std.ewubd.edu	01712345678	Student	CSE	2026-08-19	Make Admin Delete
11	Faisal Ahmed	2023-1-60-089@std.ewubd.edu	01823456789	Student	EEE	2026-08-19	Make Admin Delete
12	Zara Islam	2024-1-60-112@std.ewubd.edu	01934567890	Student	BBA	2026-08-19	Make Admin Delete
13	Karim Hassan	2024-1-60-156@std.ewubd.edu	01645678901	Student	CSE	2026-08-19	Make Admin Delete
32	Alina Khan Ayaat	2023-1-60-046@std.ewubd.edu	01712345678	Student	CSE	2026-08-19	Make Admin Delete
33	Faisal Ahmed	2023-1-60-088@std.ewubd.edu	01823456789	Student	EEE	2026-08-19	Make Admin Delete
34	Zara Islam	2024-1-60-113@std.ewubd.edu	01934567890	Student	BBA	2026-08-19	Make Admin Delete
35	Karim Hassan	2025-1-60-155@std.ewubd.edu	01645678901	Student	CSE	2026-08-19	Make Admin Delete

Figure 6.4 - Manage Users page (Manage_users.php)

Manage Categories: Add new item categories or delete existing ones and also has validation to prevent deletion if item exists.

Manage Categories — manage_categories.php

EWU Lost & Found
Manage Categories

New Category Name

Add Category

ID	Category Name	Action
1	Electronics	Delete
2	Documents	Delete
3	Bags	Delete
4	Accessories	Delete
5	Keys	Delete
6	Others	Delete

► Back to Dashboard

Figure 6.5 - Manage Categories page (Manage_categories.php)

Manage Claims: Showing all claims with claimant details and approve or reject option for pending claims.

Manage Claims — manage_claims.php

EWU Lost & Found
Manage Claims

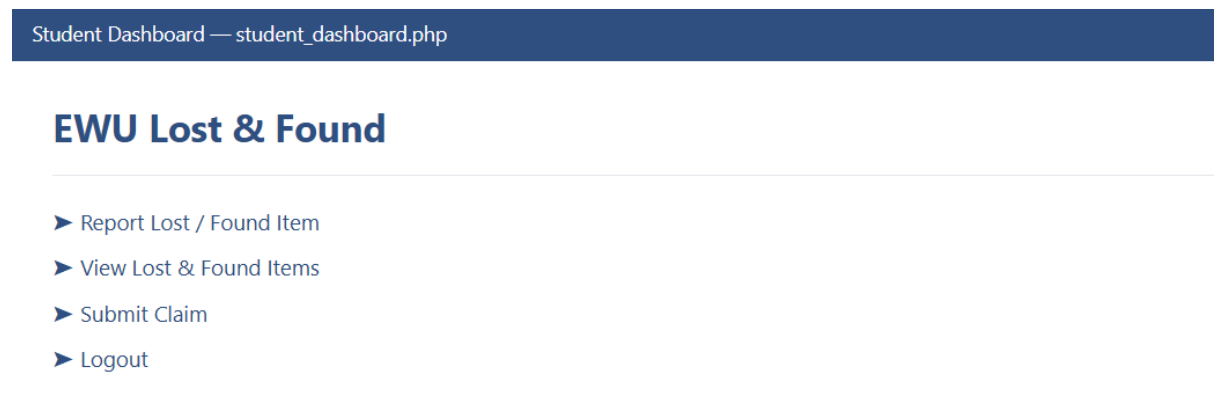
Claim ID	Item	Claimant	Claim Date	Status	Notes
38	Chemistry Textbook	Faisal Ahmed (2023-1-60-088@std.ewubd.edu)	2026-08-15	Pending	
37	Red Scarf	Tanvir Ahmed (2019-1-60-089@std.ewubd.edu)	2026-08-16	Pending	
36	Blue Earbuds	Sadia Islam (2020-2-60-102@std.ewubd.edu)	2026-08-16	Pending	
35	Blue Umbrella	Shohanur Alam Shovon (2023-1-60-271@std.ewubd.edu)	2026-08-17	Pending	
34	Gold Watch	Alina Khan Ayaat (2023-1-60-046@std.ewubd.edu)	2026-08-17	Pending	
1	Grey Backpack	Tanvir Ahmed (2019-1-60-089@std.ewubd.edu)	2026-08-15	Approved	I can identify my calculator by the sticker on the back

► Back to Dashboard

Figure 6.6 - Manage Claims page (manage_claims.php)

(Student Side)

Student Dashboard: Main menu for students which showing options to report items, view all items, submit claims and logout.



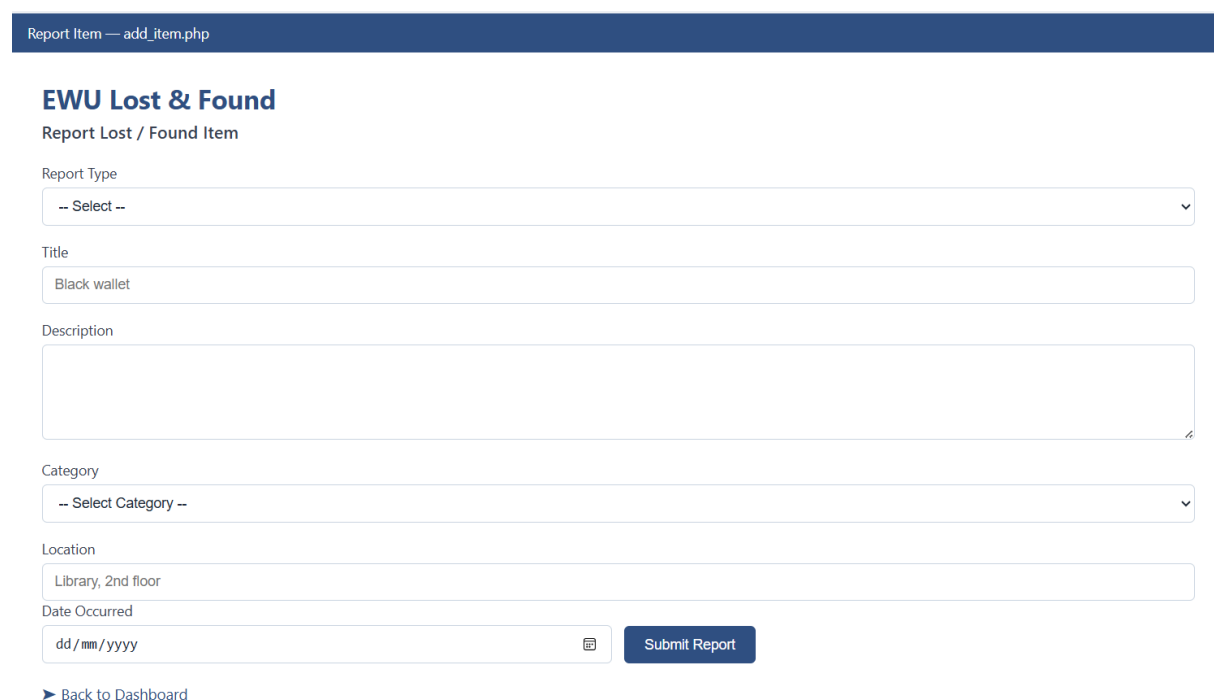
Student Dashboard — student_dashboard.php

EWU Lost & Found

- ▶ Report Lost / Found Item
- ▶ View Lost & Found Items
- ▶ Submit Claim
- ▶ Logout

Figure 6.7 - Student Dashboard (student_dashboard.php)

Report Lost/Found Item: Form to report a lost or found item with details like type, title, description, category, location and date.



Report Item — add_item.php

EWU Lost & Found

Report Lost / Found Item

Report Type
-- Select --

Title
Black wallet

Description

Category
-- Select Category --

Location
Library, 2nd floor

Date Occurred
dd/mm/yyyy

[▶ Back to Dashboard](#)

Figure 6.8 – Report Lost and Found Items(add_item.php)

View Items: Displays all reported lost or found items with status pending, approved, claimed, closed.

View Items — view_items.php

EWU Lost & Found

All Reported Items

Type	Title	Category	Location	Date Occurred	Date Reported	Status	Reported By
Lost	Chemistry Textbook	Accessories	CSE Building	2026-08-13	2026-08-19	Pending	Mim Akter
Lost	Chemistry Textbook	Accessories	CSE Building	2026-08-13	2026-08-19	Approved	Faisal Ahmed
Lost	Blue Earbuds	Bags	Library 3rd Floor	2026-08-15	2026-08-19	Approved	Sadia Islam
Lost	Blue Earbuds	Bags	Library	2026-08-15	2026-08-19	Approved	Faisal Ahmed
Found	Blue Umbrella	Documents	Main Gate	2026-08-17	2026-08-19	Pending	Shohanur Alam Shovon
Found	Red Scarf	Documents	Sports Ground	2026-08-14	2026-08-19	Approved	Tanvir Ahmed
Found	Blue Umbrella	Documents	Main Gate	2026-08-17	2026-08-19	Approved	Faisal Ahmed
Found	Red Scarf	Documents	Sports Ground	2026-08-14	2026-08-19	Approved	Faisal Ahmed
Lost	Gold Watch	Electronics	Cafeteria	2026-08-16	2026-08-19	Pending	Faisal Ahmed
Found	Silver Ring	Electronics	Mosque	2026-08-12	2026-08-19	Pending	Nur
Lost	Gold Watch	Electronics	Cafeteria	2026-08-16	2026-08-19	Approved	Faisal Ahmed

Figure 6.9 - View Lost and Found Items(view_items.php)

Submit Claim: Students select an item they want to claim and provide proof or description of ownership.

Submit Claim — add_claim.php

EWU Lost & Found

Submit a Claim

Select Item

-- Select Item --

Why is this item yours? (Proof / Description)

Submit Claim

► Back to Dashboard

Figure 6.10 – Submit Claim page (add_claims.php)

7. Conclusion

This project gave us a practical idea of how a database system works in real life. We learned how to connect tables, use PHP,HTML,CSS with MySQL, manage user roles, and keep the system secure.

During the project, we faced some challenges, especially with the Claim system and updating item status correctly. We also had to make sure that only valid EWU student emails were accepted.

In the future, the system can be expanded for EWU faculty and staff as well. The email validation can be changed to support their EWU email format(ewubd.edu) instead of only @std.ewubd.edu. We can also add notifications, search and filter options, and better security.

Overall, this project helped us understand how database concepts can be used to build a useful real-world system.

8. References

- PHP Manual-mysqli extension: <https://www.php.net/manual/en/book.mysqli.php>
- PHP Manual-password_hash() / password_verify(): <https://www.php.net/manual/en/function.password-hash.php>
- MySQL 8.0 Reference Manual: <https://dev.mysql.com/doc/refman/8.0/en/>
- W3Schools PHP & MySQL Tutorials: <https://www.w3schools.com/php/>
- XAMPP Documentation (Apache Friends): <https://www.apachefriends.org/docs/>
- draw.io Documentation: <https://www.drawio.com/doc/>